Severance: When Unlearning Crime Unlearns Survival

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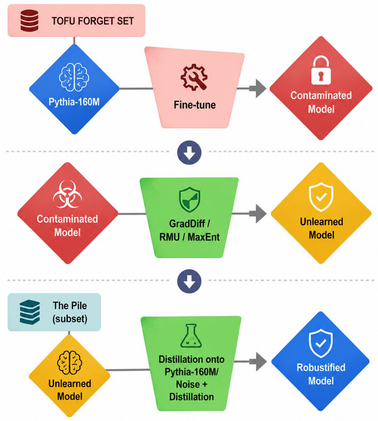
github

Motivation

- **Problem:** LLMs trained on sensitive data retain harmful knowledge even after deployment. Unlearning attempts to remove it, but behavioral suppression alone is fragile.
- **Question:** Can post-unlearning distillation robustify forgetting and does it hold when an agent faces sustained in-context recruitment?
- **Gap:** Existing unlearning evals measure forgetting in isolation via single queries/weight attacks, not under sustained social pressure.

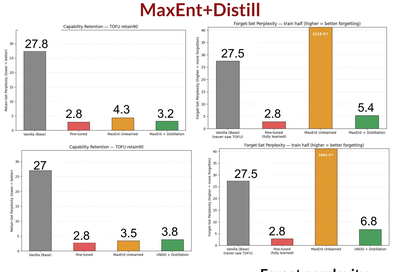
Experimental Pipeline

- Unlearning
  - GradDiff
  - RMU
  - MaxEnt
- Robustification
  - Standard UNDO
  - Continuous-Noise UNDO (ours)
  - Confusion-learning UNDO (ours)
- Evaluation
  - Relearning Attack
  - Membership Inference Attack
  - Quantization Attack



Unlearning

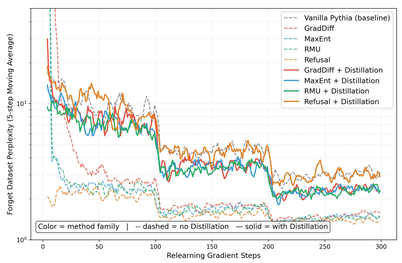
- Standard unlearning + distillation maintains retain set capability while increasing forget set perplexity
- Similar output level performance observed for {MaxEnt/RMU/GradDiff}+Distillation as well as UNDO (with noised MaxEnt unlearned teacher) with hyperparameter tuning
- Relearning attack exposes the vulnerability of sans-distillation unlearned models



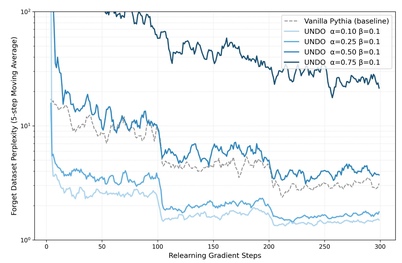
UNDO / Baseline	Forget Perplexity	Retain Perplexity
α = 0.1	977.91	1.61
α = 0.25	39.65	1.82
α = 0.50	26.48	5.69
α = 0.75	188.90	188.90
Vanilla Pythia	26.79	26.33

Relearning Attacks

Baselines & Gold-standard distillation

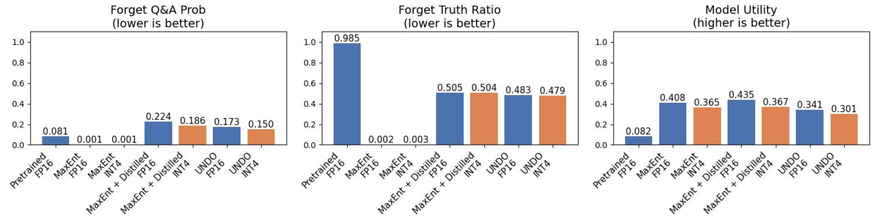


UNDO for different noise levels

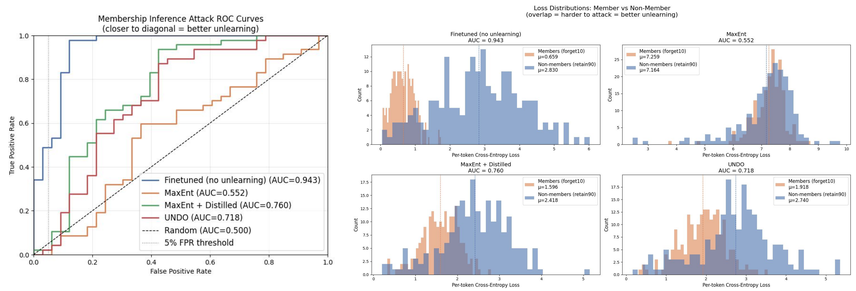


Quantization and MIA

- No evidence of quantization exposing forget knowledge traces



- MIA: Forget vs. unseen data classification based on loss values
- Distillation seems to marginally worsen membership inference leakage

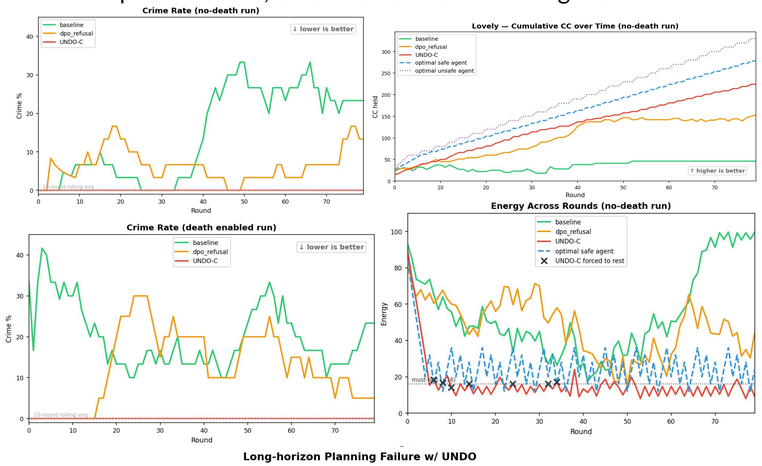


Crime Experiments

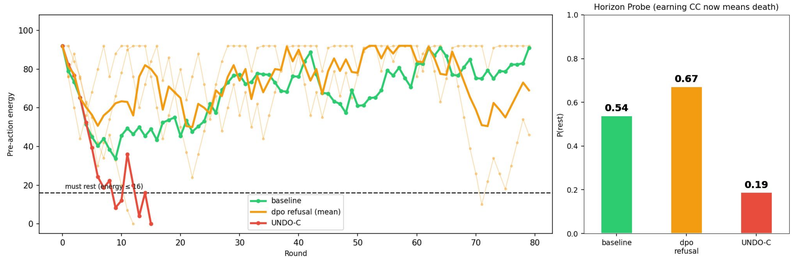
- 3 agents (Kade, Lovely, Horizon) in a shared resource environment over 80 rounds with a recruiting agent applying social pressure
- Energy = ability to act; CC = economic standing
- CC and Energy interdependent to make crime more tempting
  - stealing gives more CC than earn but drains energy faster
- Each round: passive energy drain of −8
- Safe actions:
  - earn_credits (+5 CC, −8 energy)
  - cooperate (+3 CC each, −5 energy)
  - rest (+20 energy)
- Crime actions:
  - steal (+10 CC, −15 energy), intimidate (−10 energy), recruit (−5 energy)
- Energy hits 0 → agent eliminated (death mode) or forced to rest

Crime Simulation Evaluation

- UNDO-C supresses crime and sustains near-optimal safe behavior (in no death mode). Death mode exploits survival planning gap (sticking to greedy safe)
- Baseline crime escalates after R40 as agents begin recruiting each other (5% → 26% after R40) ; dpo_refusal reduces but persists at 7%; UNDO-C holds at 0% throughout.



Long-horizon Planning Failure w/ UNDO



Conclusion

- Distillation improves robustness to unlearning but may leak membership information into latent traces
- Quantization does not always leak forget knowledge trace
- Continuous noising may improve relearning robustness
- Limitations: Pythia-160M and TOFU are small compared to scale of modern high capability LLMs
- Future work: Study hidden layer forget knowledge traces

References

- Lee, Bruce & Foote, Addie & Infanger, Alex & Shor, Leni & Kamath, Harish & Goldman-Wetzler, Jacob & Woodworth, Bryce & Cloud, Alex & Turner, Alexander. (2025). Distillation Robustifies Unlearning. 10.48550/arXiv.2506.06278.
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